Tobramycin Population PK/PD: Exploratory Pharmacodynamic and Exposure-Response Analysis

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Interpretation of Results

Key Findings

Domain	Value	Assessment
Nephrotoxicity prevalence	167/300 (55.7%)	High event rate; adequate statistical power for ER modelling
Cumulative AUC (primary exposure metric)	1232.7 ± 492.8 mg·h/L	Wide range supports logistic regression and exposure categorisation
Daily AUC_{0-24} (AUC24)	114.8 ± 38.8 mg·h/L	Consistent with 7 mg/kg q24h protocol; broad IIV warrants investigation
Trough concentration (Cmin)	0.81 ± 0.83 mg/L	Low mean trough consistent with extended-interval dosing intent

Domain	Value	Assessment
Peak concentration — central (Cmax)	26 ± 4.8 mg/L	Near clinical peak target range (25–30 mg/L) for extended-interval dosing
Exposure–response: AUC24 vs nephrotoxicity	ρ = 0.323	Moderate positive association; AUC24 is a meaningful ER predictor
Exposure–response: cumulative AUC vs nephrotoxicity	ρ = 0.326	Similar association to AUC24; cumulative exposure compounds risk
Exposure–response: Cmin vs nephrotoxicity	ρ = 0.284	Moderate association; trough-driven toxicity pathway plausible
Exposure–response: Cmax vs nephrotoxicity	ρ = 0.196	Weaker association; Cmax alone insufficient to characterise nephrotoxicity
Diabetes as nephrotoxicity risk factor	79.8% vs 43.8% (no DM)	Clinically meaningful difference; diabetes is a primary risk modifier
Hypertension as nephrotoxicity risk factor	58.2% vs 54.9% (no HTN)	Marginal difference; hypertension has limited independent contribution
CLCR effect on AUC24	ρ = -0.366	Moderate inverse correlation; CLCR drives systemic exposure
Age effect on AUC24	ρ = 0.409	Moderate positive correlation; older subjects accumulate higher AUC
Inter-individual variability in CL (ETA_1)	28.8% (approx. %CV)	Substantial unexplained IIV; covariate investigation is warranted

Nephrotoxicity Outcome

A nephrotoxicity event was recorded in **167 of 300 subjects (55.7%)**. This event rate is higher than the 10–25% typically reported in general clinical populations receiving standard aminoglycoside dosing, which likely reflects both the extended-interval protocol and the high prevalence of nephrotoxicity risk factors in this cohort (diabetes mellitus 33.0%, prior tobramycin courses in 84.3% of subjects). The elevated event rate is, however, favourable from a statistical standpoint: it provides adequate power for logistic regression-based exposure-response modelling without the need for outcome rebalancing.

Exposure Metrics and Inter-Individual Variability

Five PK-derived exposure metrics were characterised at the subject level (Table 2). The mean steady-state $AUC_{0\text{–}24}$ was **114.8 ± 38.8 mg·h/L** (range 49–257), representing a 5.2-fold range across subjects. Cumulative AUC over the treatment course ranged from 491.3 to 3377.2 mg·h/L — a nearly 7-fold range — underscoring that the overall nephrotoxic burden is not well-captured by a single-day exposure metric alone.

The mean central-compartment Cmax of **26 ± 4.8 mg/L** is consistent with the clinical efficacy target range of 25–30 mg/L for extended-interval tobramycin dosing, although individual values ranged from 17.6 to 47.1 mg/L. Mean trough concentration (Cmin) was **0.81 ± 0.83 mg/L**, with the majority of subjects achieving near-zero troughs as intended under extended-interval dosing to minimise nephrotoxic exposure during the drug-free interval.

Exposure-Response Relationships

All five PK metrics showed positive Spearman correlations with the binary nephrotoxicity outcome, with the strongest associations observed for **cumulative AUC** (ρ = 0.326) and **AUC24** (ρ = 0.323). Subjects who developed nephrotoxicity had a mean cumulative AUC of 1387.6 mg·h/L, compared with 1038.2 mg·h/L in those who did not (34% higher exposure). A similar pattern was observed for AUC24: 126.8 vs 99.7 mg·h/L.

The Cmin correlation (ρ = 0.284) is of clinical relevance: persistent trough concentrations above 0.5–1 mg/L have been linked to proximal tubular toxicity in prior aminoglycoside studies, and the observed Cmin range (up to 4.8 mg/L) suggests a subset of subjects with incomplete drug elimination between doses. Cmax (ρ = 0.196) showed the weakest association, consistent with the current understanding that aminoglycoside nephrotoxicity is driven predominantly by cumulative and trough exposure rather than peak concentration.

Quartile analyses of exposure metrics (see Exposure-Response section) confirmed a monotonically increasing nephrotoxicity rate with increasing cumulative AUC and AUC24 quartile, supporting a causal interpretation of the AUC-nephrotoxicity relationship.

Covariate Effects on Pharmacokinetics

Among continuous covariates, **age** showed the strongest positive correlation with AUC24 (ρ = 0.409), consistent with the age-associated decline in renal function and the resulting reduction in tobramycin clearance. **CLCR** showed a moderate inverse correlation with AUC24 (ρ = -0.366) and cumulative AUC (ρ = -0.392), as expected for a renally cleared drug: subjects with lower CLCR accumulate higher systemic exposure per dose. Body weight showed a weaker positive correlation with AUC24 (ρ = 0.203), likely reflecting the increase in volume of distribution in heavier subjects in the context of weight-based dosing.

Notably, the four inter-individual variability terms (ETAs) for CL, V1, Q, and V2 showed minimal correlation with any covariate in the pairwise analysis (|ρ| < 0.15 in all cases), suggesting that the base PopPK model was estimated **without covariate terms included**. This residual ETA-covariate structure constitutes the primary justification for covariate model building: the substantial IIV in CL (~29% CV) that remains unexplained by the base model is partially attributable to identifiable patient characteristics (CLCR, age) and should be reduced through formal covariate inclusion.

Covariate Effects on Nephrotoxicity Risk

Diabetes mellitus was the most clinically important risk modifier: the nephrotoxicity rate was 79.8% in diabetic subjects versus 43.8% in non-diabetic subjects — a **36-percentage-point absolute difference** that warrants inclusion in any exposure-response model as a binary effect modifier. Hypertension showed minimal independent effect (58.2% vs 54.9%), suggesting that its apparent association may be confounded by co-occurrence with diabetes or age.

Increasing age and declining CLCR were both associated with higher nephrotoxicity rates in the continuous covariate analysis, consistent with the mechanism: these factors reduce drug clearance, increase systemic and tubular exposure, and lower the renal reserve available to withstand nephrotoxic insult.

Recommendations

1. **Use AUC24 or cumulative AUC as the primary PK metric in the exposure-response model.** Both metrics demonstrated moderate positive associations with nephrotoxicity ($\rho \approx 0.32$–0.33) and are mechanistically justified for an AUC-driven toxicity outcome. Cumulative AUC should be preferred when characterising the total nephrotoxic burden across the full treatment course.

2. **Incorporate Cmin as a secondary exposure metric.** The Cmin association ($\rho = 0.284$) supports evaluation of a dual AUC + Cmin predictor or a logistic regression model that includes both variables as independent covariates, reflecting different nephrotoxic mechanisms.

3. **Include diabetes mellitus as a binary covariate in the exposure-response model.** The 79.8% nephrotoxicity rate in diabetic subjects versus 43.8% in non-diabetic subjects represents the largest single risk modifier identified in this analysis and should be treated as a mandatory covariate in any logistic or time-to-event ER model.

4. **Incorporate CLCR as a covariate on CL in the structural PK model.** The inverse relationship ($\rho = -0.366$) confirms the physiological expectation and will reduce unexplained IIV in CL. Age should be evaluated as a secondary covariate or as a surrogate for CLCR decline.

5. **Apply allometric or empirical weight scaling on volume of distribution terms.** The positive weight-AUC correlation ($\rho = 0.203$), combined with the protocol-defined 7 mg/kg dosing, supports allometric scaling of V1 and V2 with body weight to reduce residual IIV.

6. **Evaluate non-linear (sigmoidal) exposure-response functions.** The quartile plots suggest a non-linear, steepening nephrotoxicity rate with increasing exposure. An Emax or logistic Hill function may provide a better fit than a linear logistic model, particularly for high-AUC subjects.

7. **Investigate the basis for substantial unexplained IIV in Q and V2.** The peripheral compartment parameters (eta_q SD ≈ 25%, eta_v2 SD ≈ 20%) showed no meaningful covariate associations, suggesting that distributional heterogeneity may reflect unmeasured biological factors (tissue binding, hydration status) rather than the covariates available in this dataset.

PK Metric Summary Statistics

PK Metric	N	Missing (%)	Mean	Median	SD	Min	Max
Area under the concentration-time curve over 24 hours (mg*h/L)	300	0	114.759	110.108	38.750	48.959	257.012
Predicted or estimated trough concentration (minimum concentration, mg/L)	300	0	0.814	0.547	0.825	0.011	4.804
Cumulative exposure over treatment course (sum of AUC values, mg*h/L)	300	0	1,232.661	1,156.508	492.797	491.271	3,377.231
Maximum concentration in the central compartment (mg/L)	300	0	26.048	25.149	4.831	17.615	47.113
Maximum concentration in the peripheral compartment (mg/L)	300	0	9.944	9.429	2.688	5.096	20.343

All units: AUC24 and cumulative AUC in mg·h/L; concentrations (Cmin, Cmax) in mg/L.

ETA vs Covariates Visualisations

The following plots display the distribution of individual random effects (ETAs) from the base population PK model against subject-level covariates. Scatter plots with Spearman ρ are used for continuous covariates; box-and-whisker plots are used for categorical covariates. A horizontal reference line at zero is included in all panels.

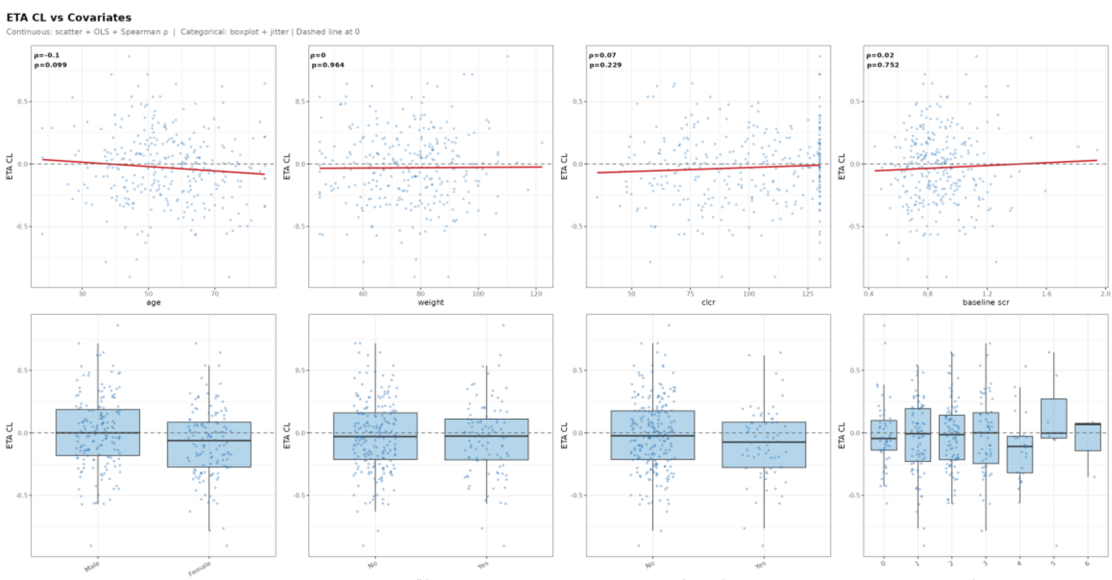

Figure 1. Individual random effect for clearance (ETA_1, CL) versus subject-level covariates.

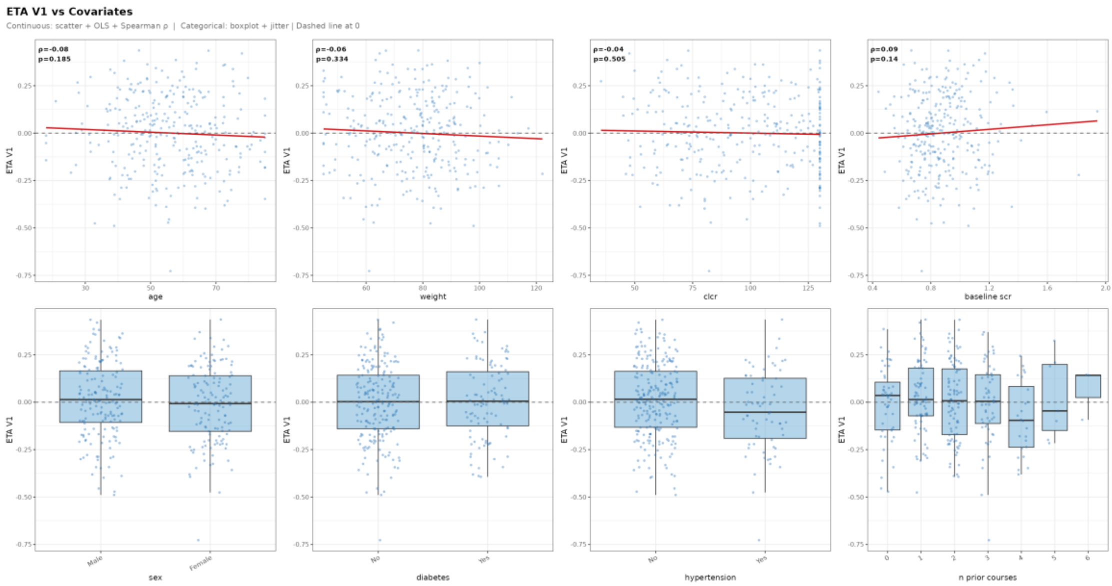

Figure 2. Individual random effect for central volume of distribution (ETA_2, V1) versus subject-level covariates.

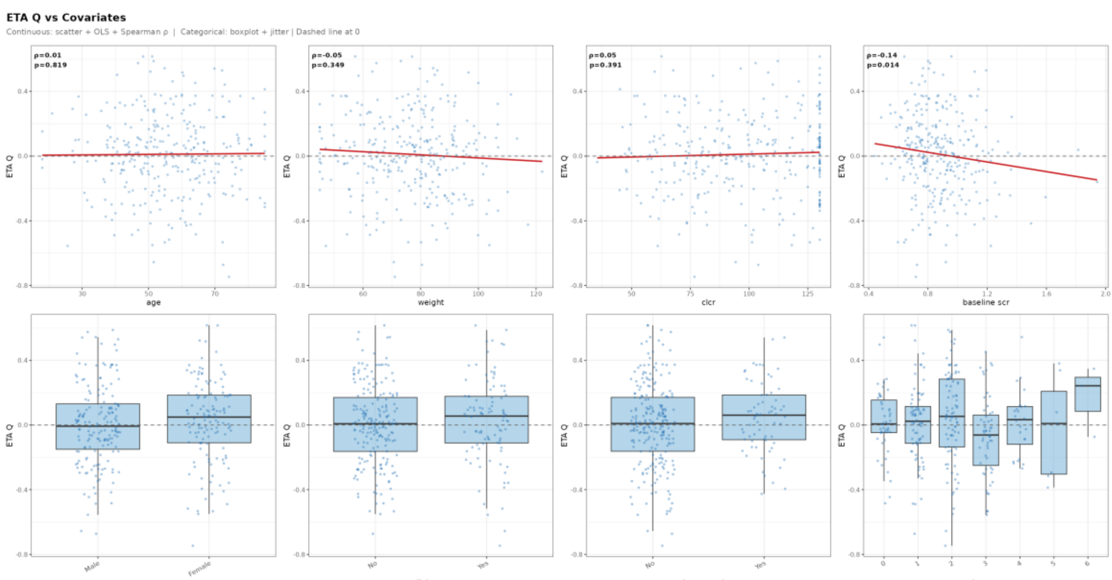

Figure 3. Individual random effect for inter-compartmental clearance (ETA_3, Q) versus subject-level covariates.

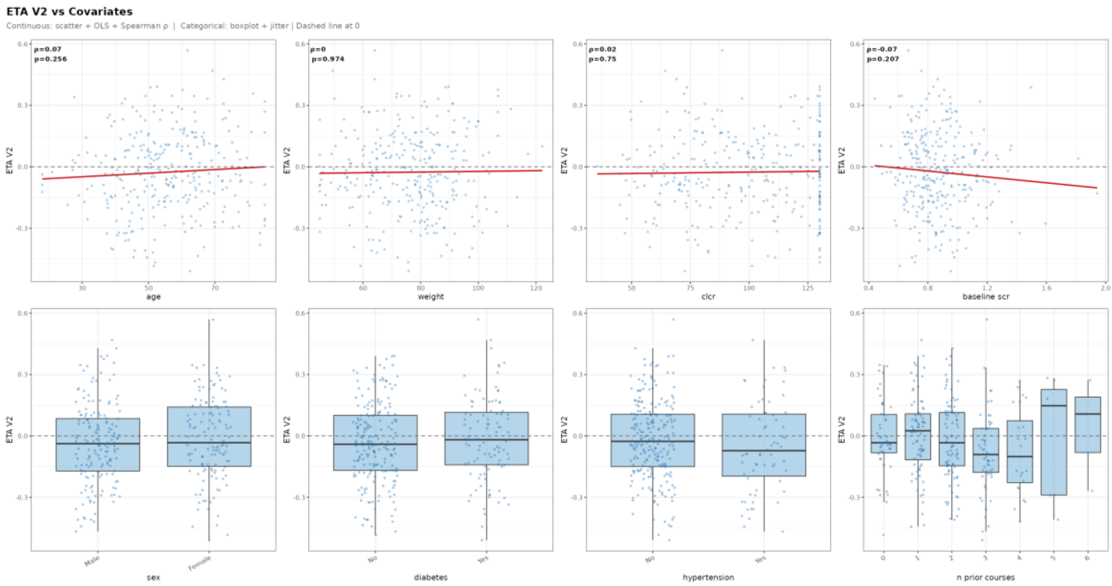

Figure 4. Individual random effect for peripheral volume of distribution (ETA_4, V2) versus subject-level covariates.

PK Metrics vs Covariates Visualisations

The following plots display each subject-level PK exposure metric against all available covariates. Scatter plots with Spearman ρ are used for continuous covariates; box-and-whisker plots are used for categorical covariates.

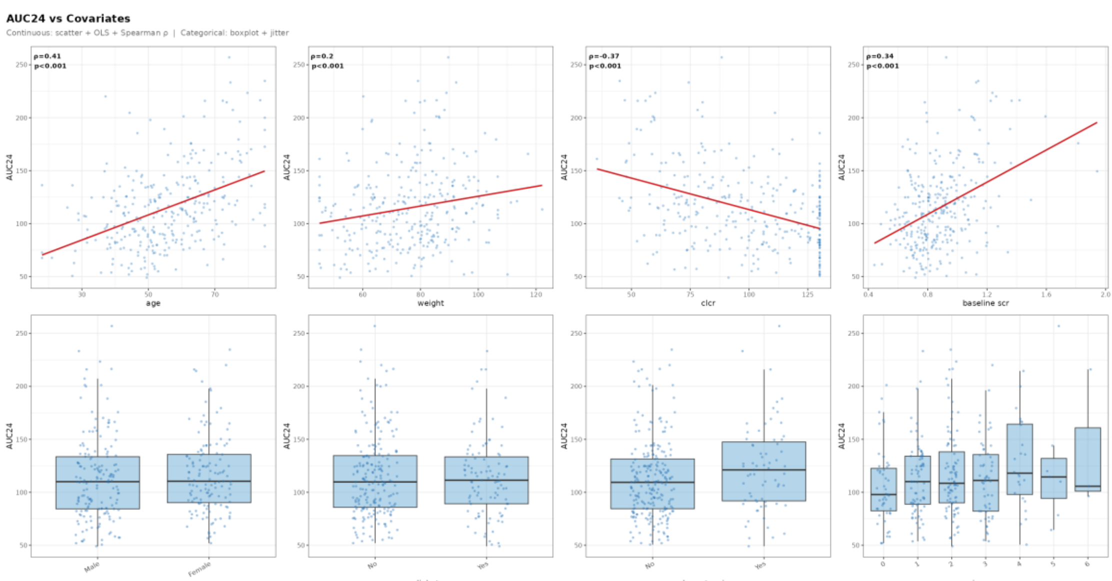

Figure 5. Daily AUC (AUC24, mg·h/L) versus subject-level covariates.

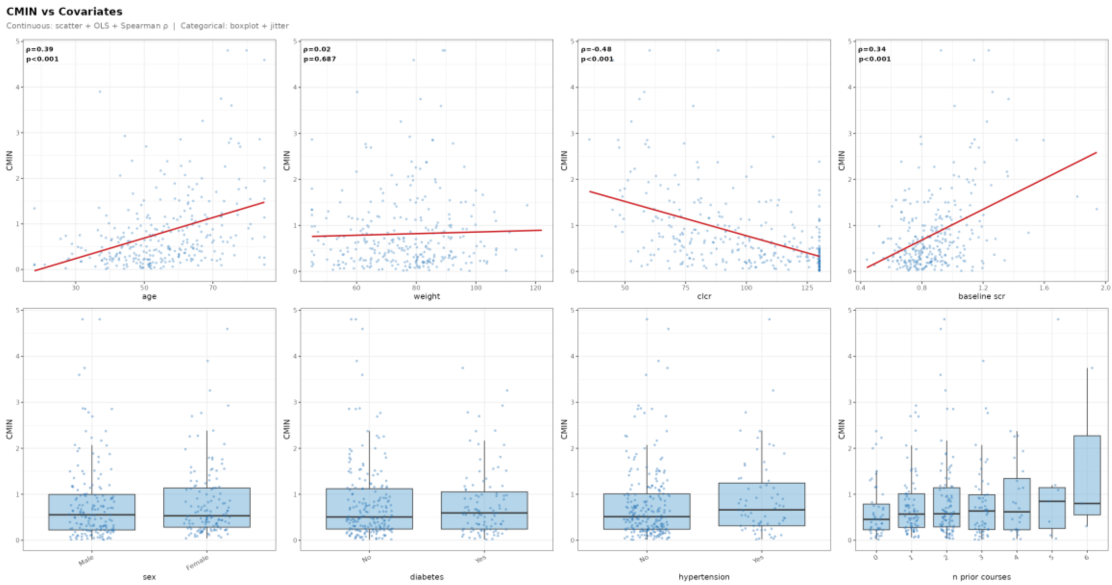

Figure 6. Trough concentration (Cmin, mg/L) versus subject-level covariates.

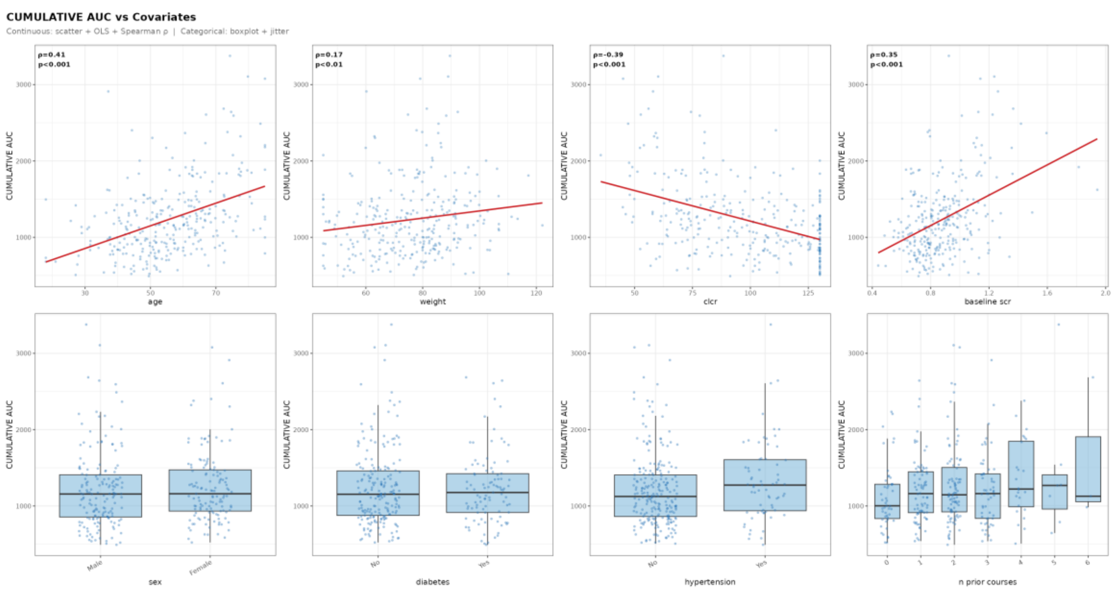

Figure 7. Cumulative AUC (mg·h/L) versus subject-level covariates.

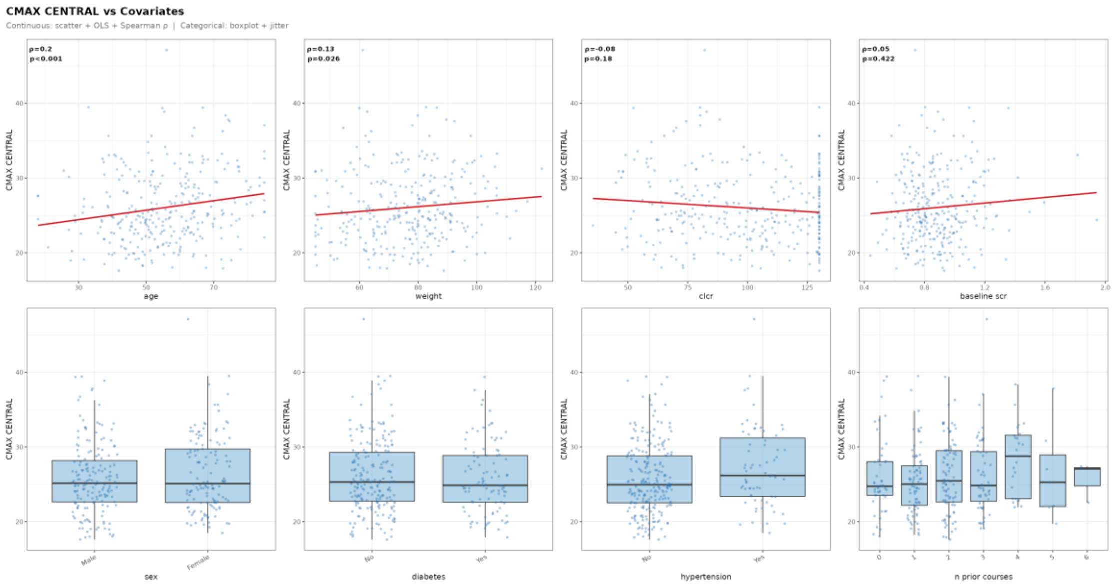

Figure 8. Peak concentration in the central compartment ($Cmax_1$, mg/L) versus subject-level covariates.

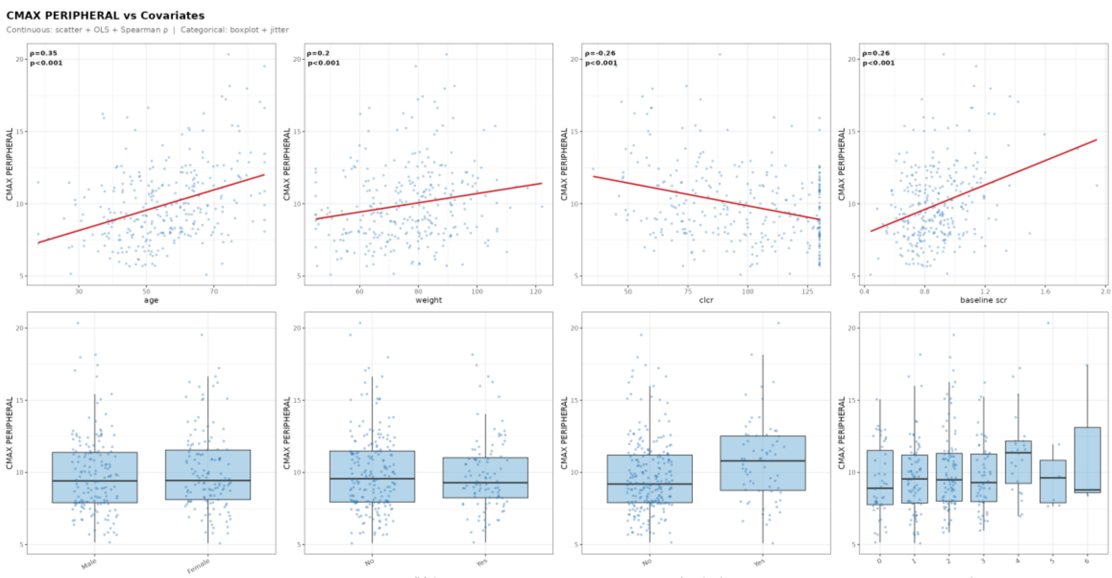

Figure 9. Peak concentration in the peripheral compartment ($Cmax_2$, mg/L) versus subject-level covariates.

Response vs Covariates Visualisations

The following plots display the nephrotoxicity event rate (proportion of subjects with nephro_binary = 1) stratified by covariate. For continuous covariates, subjects are grouped into quartiles; for categorical covariates, proportions are shown per category level.

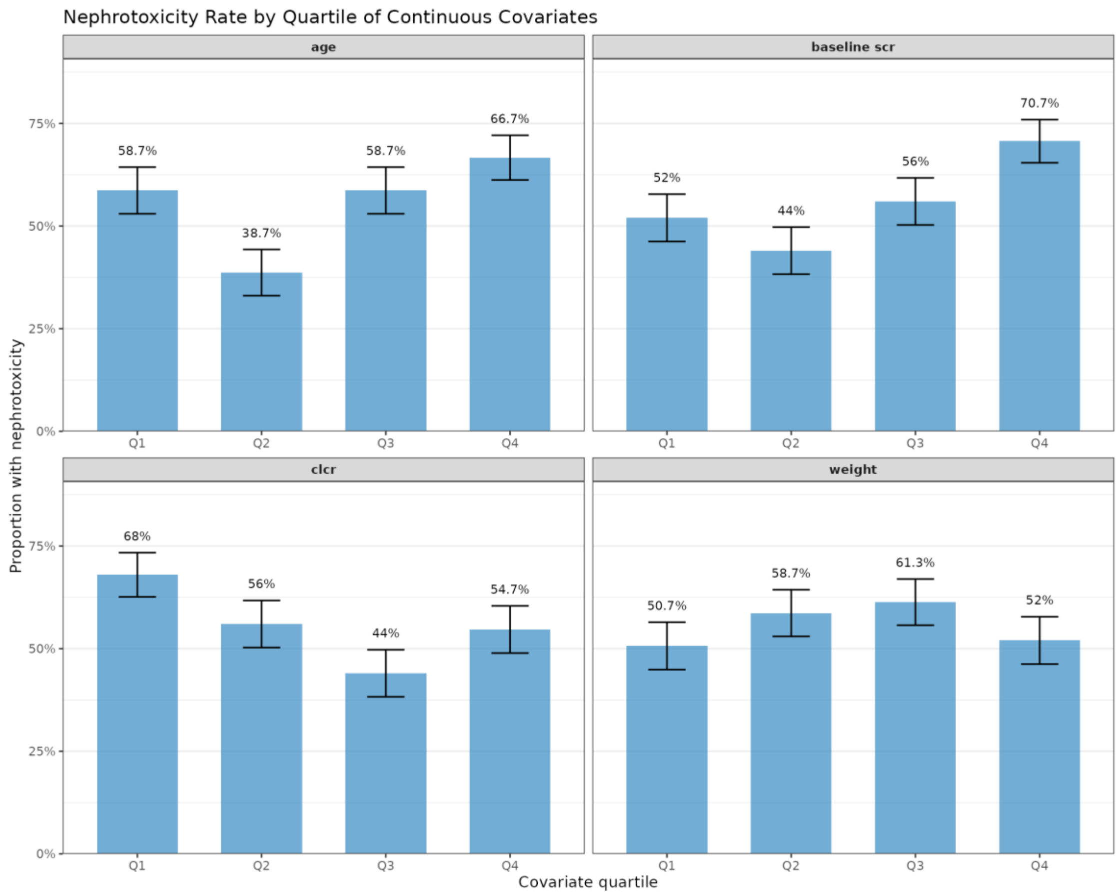

Figure 10. Nephrotoxicity rate by quartile of continuous covariate. Error bars denote ±1 standard error of the proportion.

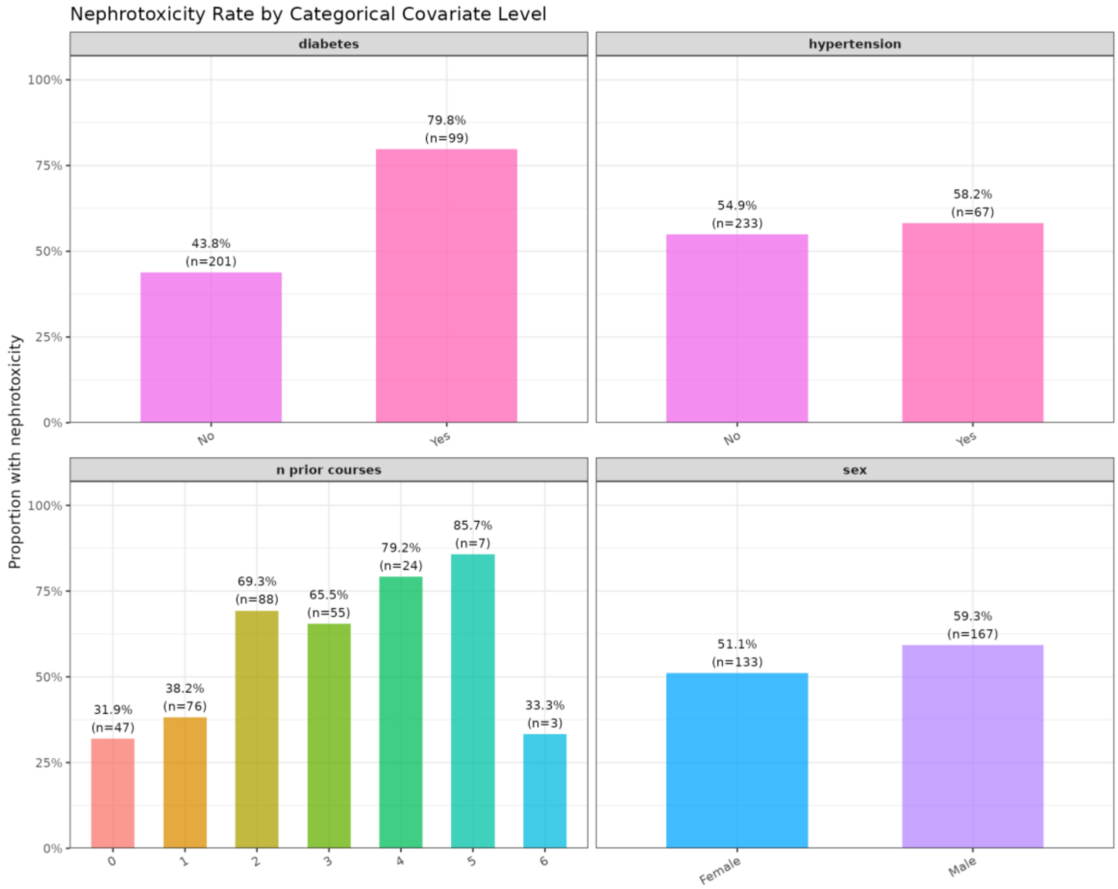

Figure 11. Nephrotoxicity rate by categorical covariate level.

Exposure-Response Visualisations

The following plots characterise the relationship between PK exposure metrics and the binary nephrotoxicity outcome. Box plots display the distribution of each PK metric stratified by nephrotoxicity status; quartile plots show the monotonic trend in nephrotoxicity rate with increasing exposure.

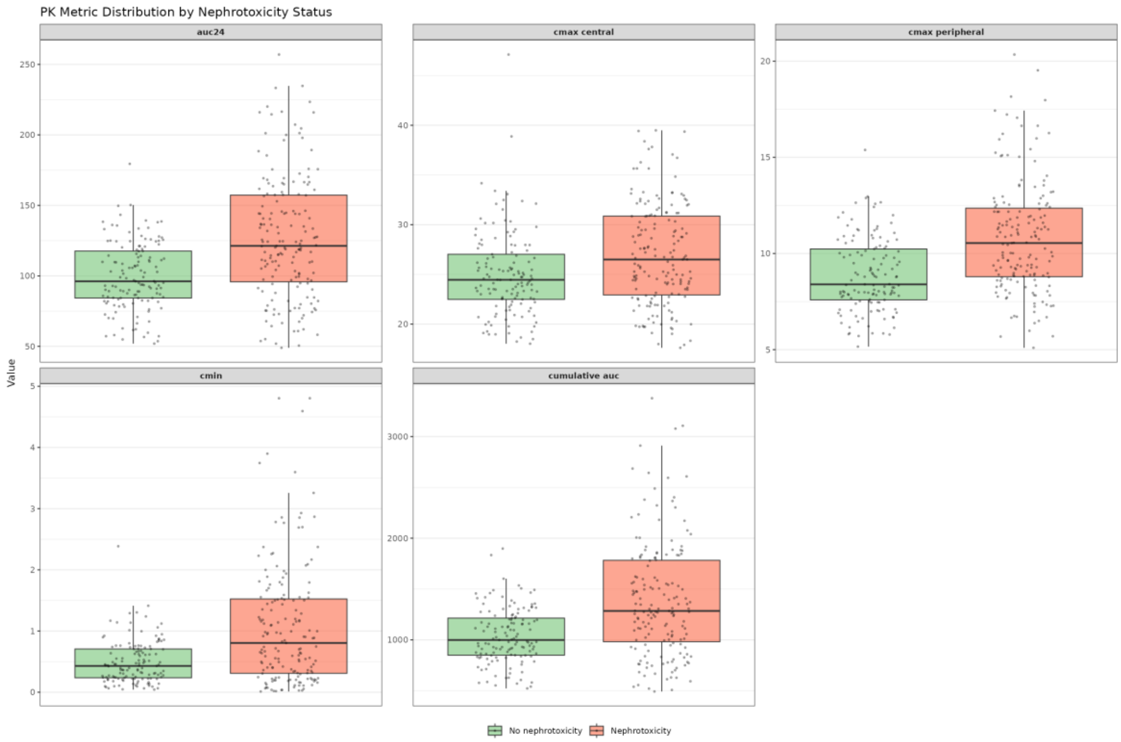

Figure 12. Distribution of PK exposure metrics stratified by nephrotoxicity outcome. Individual subject observations are overlaid (jitter).

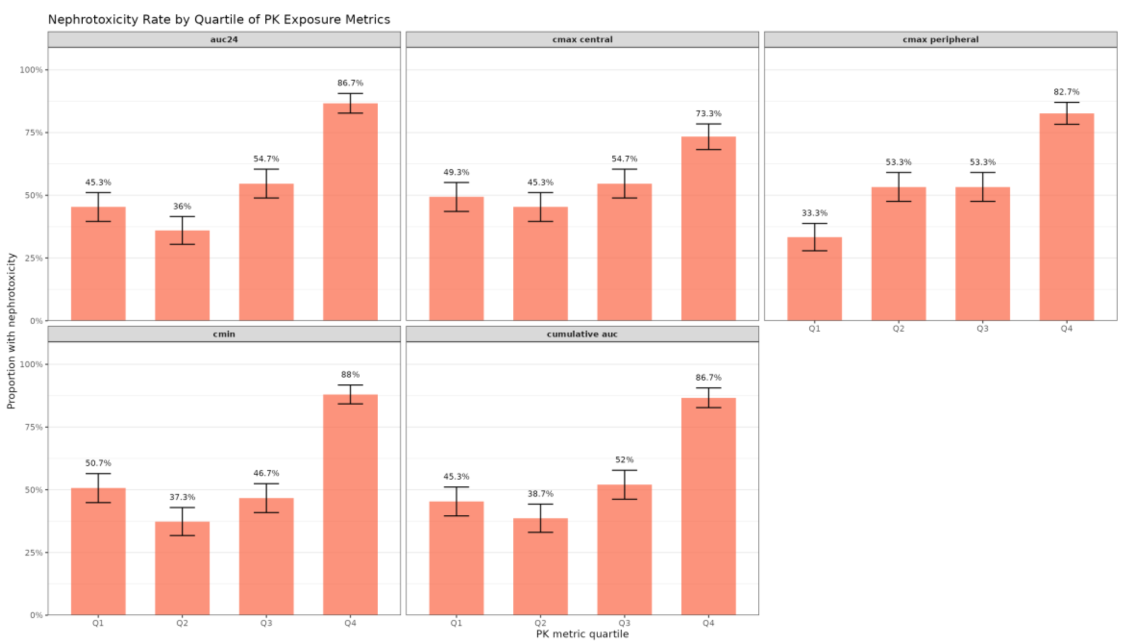

Figure 13. Nephrotoxicity rate by quartile of each PK exposure metric. Error bars denote ±1 standard error of the proportion.